

Pearson Edexcel International GCSE

Mathematics A

Modular Unit 1

January 2022 Reconstructed Past Paper

DETAILED SOLUTIONS

Original question followed by a full worked solution

27 adaptive question sections 76 marks

Selected from Linear Higher Papers 1H and 2H

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L-to-M | CONVERTED MODULAR EDITION

Selected from Linear Higher papers; not an original Modular examination paper.

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About this reconstructed paper

Each original source section is followed by the demand-local candidate solution and final answer in the same order as the Student book.

Ownership rule: Unit 2 owns a demand when its assessed or terminal statement is Unit 2, even if Unit 1 skills are prerequisites. This book contains the demands owned by Unit 1; the selection contains 76 marks.

This is an Elite reconstructed teaching paper selected from official Linear Higher material; it is not an official Pearson Modular examination paper. Candidate source assets are retained for private teaching use only.

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Original question 01

4MA1-1H Q1 Q1(a)

ORIGINAL QUESTION**1 marks****1** (a) Simplify $a^7 \times a^4$

(1)

Worked solution 01

Chapter: Algebraic Roots & Indices

MODULAR UNIT 1

1 marks

Q1(a) – Chapter: Algebraic Roots & Indices

Family: Combine indexed algebraic factors • Variant: Multiply or divide common bases

Method

$$a^7 \times a^4 = a^{(7+4)} = a^{11}.$$

*Why: When multiplying powers with the same base, add the indices.***Final answer****simplified form:** a^{11}

Original question 02

4MA1-1H Q1 Q1(b)

ORIGINAL QUESTION**1 marks**(b) Simplify $w^{15} \div w^3$

(1)

Worked solution 02

Chapter: Algebraic Roots & Indices

MODULAR UNIT 1

1 marks

Q1(b) – Chapter: Algebraic Roots & Indices

Family: Combine indexed algebraic factors • Variant: Multiply or divide common bases

Method

$$w^{15} \div w^3 = w^{(15-3)} = w^{12}.$$

*Why: When dividing powers with the same base, subtract the indices.***Final answer****simplified form:** w^{12}

Original question 03

4MA1-1H Q1 Q1(c)

ORIGINAL QUESTION**2 marks**(c) Simplify $(8x^5y^3)^2$

(2)

Worked solution 03

Chapter: Algebraic Roots & Indices

MODULAR UNIT 1

2 marks

Q1(c) – Chapter: Algebraic Roots & Indices

Family: Combine indexed algebraic factors • Variant: Power of a product or power

Method

$$(8x^5y^3)^2 = 8^2 x^{10} y^6.$$

*Why: Square the coefficient and multiply each exponent by 2.***Method**

$$\text{Therefore } (8x^5y^3)^2 = 64x^{10}y^6.$$

*Why: $8^2 = 64$.***Final answer****simplified form:** $64x^{10}y^6$

Original question 04

4MA1-1H Q3

ORIGINAL QUESTION**3 marks**

- 3 The diagram shows triangle ABC

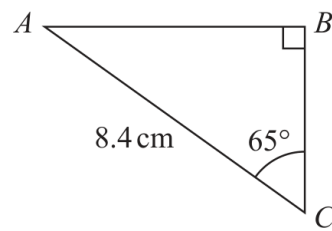


Diagram **NOT**
accurately drawn

Work out the length of the side AB
Give your answer correct to 3 significant figures.

..... cm

Worked solution 04

Chapter: Right-Angled Triangles - Pythagoras & Trigonometry

MODULAR UNIT 1

3 marks

Q3 – Chapter: Right-Angled Triangles - Pythagoras & Trigonometry

Family: Use right-triangle trigonometry for lengths • Variant: Find opposite side with sine or tangent

Method

$$AB/8.4 = \sin 65^\circ / \sin 90^\circ.$$

*Why: Use the sine rule in the right triangle.***Method**

$$AB = 8.4 \sin 65^\circ.$$

*Why: $\sin 90^\circ = 1$, so rearrange for AB.***Method**

$$AB = 7.61 \text{ cm (3 s.f.)}.$$

*Why: Round the calculator value 7.612... to 3 significant figures.***Final answer****AB: 7.61 cm**

Original question 05

4MA1-1H Q6 Q6(b)(i), Q6(b)(ii)

ORIGINAL QUESTION**3 marks**(b) (i) Factorise $y^2 - 2y - 35$

(2)(ii) Hence, solve $y^2 - 2y - 35 = 0$

(1)

Worked solution 05

Chapter: Factorising; Solving Quadratic Equations

MODULAR UNIT 1

3 marks

Q6(b)(i) – Chapter: Factorising

Family: Factorise a non-monic or special quadratic • Variant: Factorise as the first part of a factor-then-solve chain

Method

We need two numbers with product -35 and sum -2 : -7 and $+5$.

Why: The constant term is -35 and the x -coefficient is -2 .

Method

$$y^2 - 2y - 35 = (y - 7)(y + 5).$$

Why: Expand the brackets to check.

Final answer

factorisation: $(y - 7)(y + 5)$

Q6(b)(ii) – Chapter: Solving Quadratic Equations

Family: Solve a quadratic by factorisation • Variant: Monic quadratic roots by factors

Method

$$(y - 7)(y + 5) = 0, \text{ so } y = 7 \text{ or } y = -5.$$

Why: Use the zero-product rule; this follows the factorisation in part (i).

Final answer

roots: $y = 7$ or $y = -5$

Original question 06

4MA1-1H Q7

ORIGINAL QUESTION**3 marks**

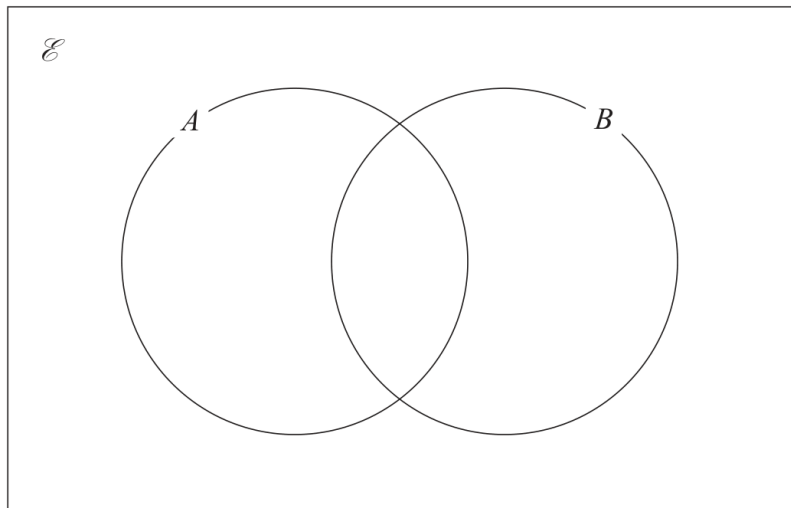
7 $\mathcal{E} = \{4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15\}$

$$A \cap B = \{5, 10, 15\}$$

$$B' = \{7, 8, 9, 11, 12, 13, 14\}$$

$$A' = \{4, 6, 7, 8, 14\}$$

Complete the Venn diagram for this information.



Worked solution 06

Chapter: Set Notation & Venn Diagrams

MODULAR UNIT 1

3 marks

Q7 – Chapter: Set Notation & Venn Diagrams

Family: Read or construct a two-set Venn diagram • Variant: Complete a two-set diagram from membership data

Method

$$A \cap B = \{5, 10, 15\}.$$

Why: These elements belong to both sets.

Method

$$A \text{ only} = \{9, 11, 12, 13\}, B \text{ only} = \{4, 6\}, \text{ and outside} = \{7, 8, 14\}.$$

Why: Use the complements in the given universal set.

Method

Place the three groups in the corresponding Venn regions.

Why: Each element is used exactly once.

Final answer

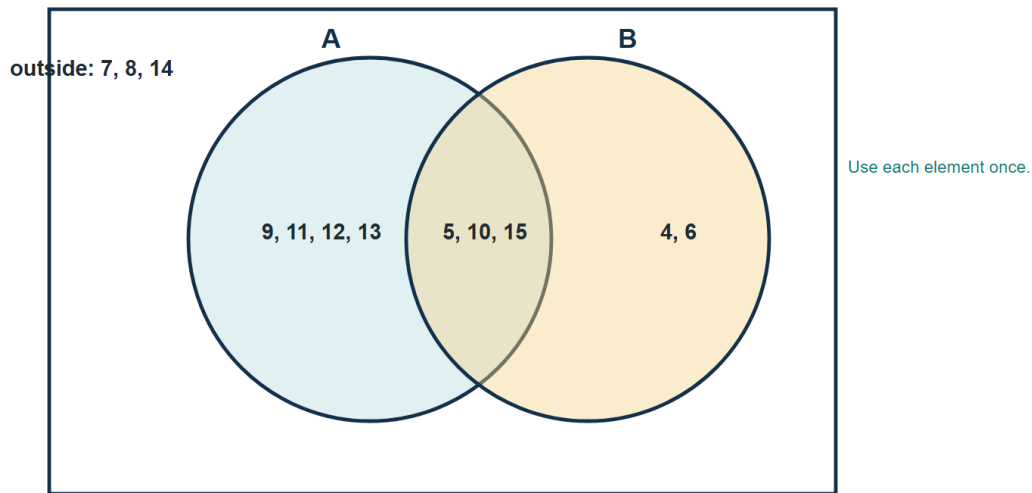
Venn regions: A only $\{9, 11, 12, 13\}$; $A \cap B$ $\{5, 10, 15\}$; B only $\{4, 6\}$; outside $\{7, 8, 14\}$

Completed diagram 06

Chapter: Set Notation & Venn Diagrams

MODULAR UNIT 1

3 marks

Q7 • Completed Venn diagram $A \cap B = \{5, 10, 15\}$; A only = $\{9, 11, 12, 13\}$; B only = $\{4, 6\}$ 

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Original question 07

4MA1-1H Q9

ORIGINAL QUESTION**4 marks**

- 9 The diagram shows isosceles triangle ABC

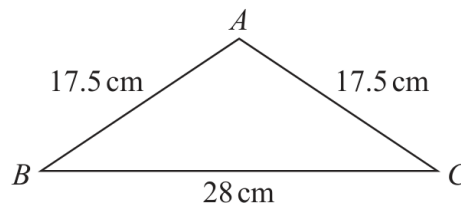


Diagram **NOT**
accurately drawn

$$AB = AC = 17.5 \text{ cm}$$

$$BC = 28 \text{ cm}$$

Calculate the area of triangle ABC

..... cm^2

Worked solution 07

Chapter: Area & Perimeter

MODULAR UNIT 1

4 marks

Q9 – Chapter: Area & Perimeter

Family: Use parallelogram and trapezium area • Variant: Combine polygon area with Pythagoras or trigonometry

Method

The perpendicular height is $\sqrt{(17.5^2 - 14^2)} = \sqrt{110.25} = 10.5$ cm.

Why: Bisect the 28 cm base of the isosceles triangle and use Pythagoras.

Method

Area = $\frac{1}{2} \times 28 \times 10.5$.

Why: Use base times perpendicular height divided by 2.

Method

Area = 147 cm^2 .

Why: Evaluate the product.

Method

Therefore the area is 147 cm^2 .

Why: State the requested area.

Final answer

area: 147 cm^2

Original question 08

4MA1-1H Q10 Q10(a)

ORIGINAL QUESTION**2 marks**

10 The straight line **L** has equation $2y + 7x = 10$

(a) Find the gradient of **L**

.....
(2)

Worked solution 08Chapter: Linear Graphs ($y = mx + c$)

MODULAR UNIT 1

2 marks

Q10(a) – Chapter: Linear Graphs ($y = mx + c$)Family: Read gradient and intercept structure • Variant: Gradient and intercept from $y=mx+c$ **Method** $2y+7x=10$ gives $2y=10-7x$.*Why: Rearrange into $y=mx+c$ form.***Method** $y=5-3.5x$, so the gradient is -3.5 .*Why: The coefficient of x is the gradient.***Final answer**gradient: -3.5

Original question 09

4MA1-1H Q10 Q10(b)

ORIGINAL QUESTION**1 marks**

10 The straight line **L** has equation $2y + 7x = 10$

(a) Find the gradient of **L**

(b) Find the coordinates of the point where **L** crosses the y -axis.

(.....,)
(1)

Worked solution 09Chapter: Linear Graphs ($y = mx + c$)

MODULAR UNIT 1

1 marks

Q10(b) – Chapter: Linear Graphs ($y = mx + c$)Family: Read gradient and intercept structure • Variant: Gradient and intercept from $y=mx+c$ **Method**

At the y-axis $x=0$, so $2y=10$ and $y=5$; the intercept is $(0,5)$.

Why: Set x equal to zero to find the y-intercept.

Final answer

y-intercept: $(0,5)$

Original question 10

4MA1-1H Q13 Q13(a)

ORIGINAL QUESTION**3 marks****13** (a) Expand and simplify $5x(x + 2)(3x - 4)$

3

(3)

Worked solution 10

Chapter: Expanding Brackets

MODULAR UNIT 1

3 marks

Q13(a) – Chapter: Expanding Brackets

Family: Expand three or more linear factors • Variant: Monomial times two linear brackets

Method

$$5x(x+2)=5x^2+10x.$$

*Why: Multiply the first two factors.***Method**

$$(5x^2+10x)(3x-4)=15x^3-20x^2+30x^2-40x.$$

*Why: Multiply by the third factor.***Method**Collecting terms gives $15x^3+10x^2-40x$.*Why: Combine the two x^2 terms.***Final answer**expanded expression: $15x^3+10x^2-40x$

Original question 11

4MA1-1H Q13 Q13(b)

ORIGINAL QUESTION**3 marks**

(b) Simplify completely $\left(\frac{16w^8}{y^{20}}\right)^{-\frac{3}{4}}$

(3)

Worked solution 11

Chapter: Algebraic Roots & Indices

MODULAR UNIT 1

3 marks

Q13(b) – Chapter: Algebraic Roots & Indices

Family: Combine indexed algebraic factors • Variant: Complex indexed monomial or fractional outer power

Method

$$(16w^8/y^{20})^{(-3/4)} = y^{15}/(16^{(3/4)}w^6).$$

Why: Apply the $-3/4$ power to each factor and invert.

Method

$$16^{(3/4)}=8 \text{ and } w^{(-6)}=1/w^6.$$

Why: Evaluate the coefficient and rewrite the negative exponent.

Method

The simplified result is $y^{15}/(8w^6)$.

Why: Combine the factors.

Final answer

simplified expression: $y^{15}/(8w^6)$

Original question 12

4MA1-1H Q14 Q14(a)

ORIGINAL QUESTION

2 marks

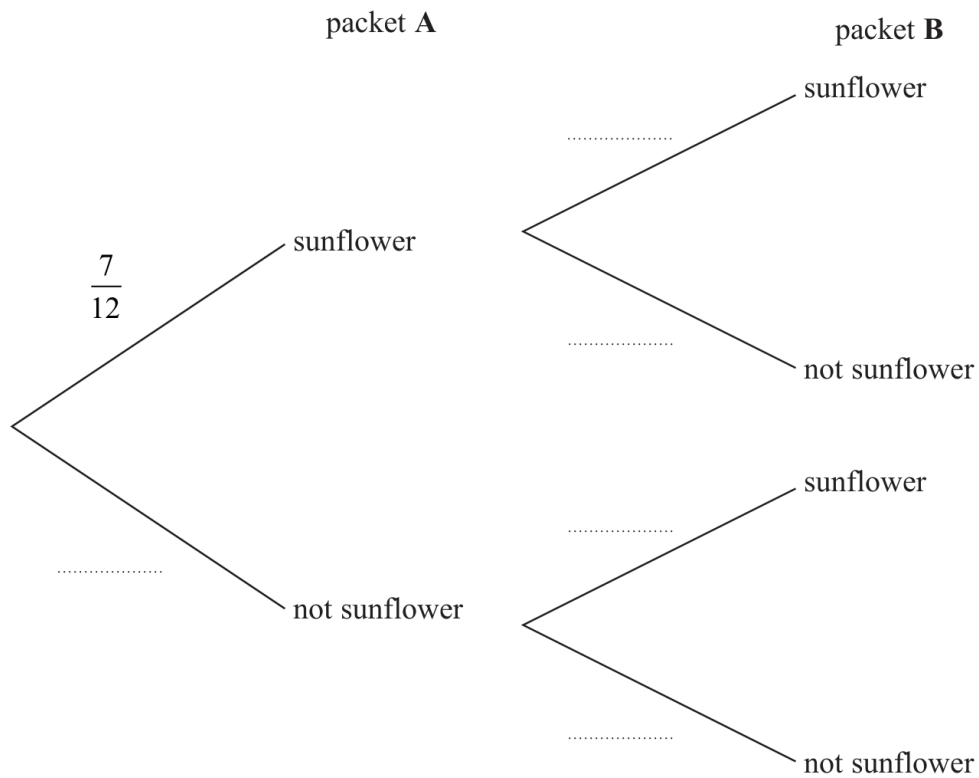
14 Aika has 2 packets of seeds, packet **A** and packet **B**

There are 12 seeds in packet A and 7 of these are sunflower seeds.

There are 15 seeds in packet **B** and 8 of these are sunflower seeds.

Aika is going to take at random a seed from packet **A** and a seed from packet **B**

(a) Complete the probability tree diagram.



(2)

Worked solution 12

Chapter: Probability Diagrams - Venn & Tree Diagrams

MODULAR UNIT 1

2 marks

Q14(a) – Chapter: Probability Diagrams - Venn & Tree Diagrams

Family: Construct and read probability trees • Variant: Complete an independent-event tree

Method

From packet A the branches are sunflower $7/12$ and not sunflower $5/12$; from packet B they are sunflower $8/15$ and not sunflower $7/15$.

Why: Use the given seed counts to complete all four branches.

Method

Place these four probabilities on the supplied tree.

Why: The second draw is from a different packet, so the branch probabilities remain unchanged.

Final answer

tree probabilities: $7/12$, $5/12$, $8/15$, $7/15$

Completed diagram 12

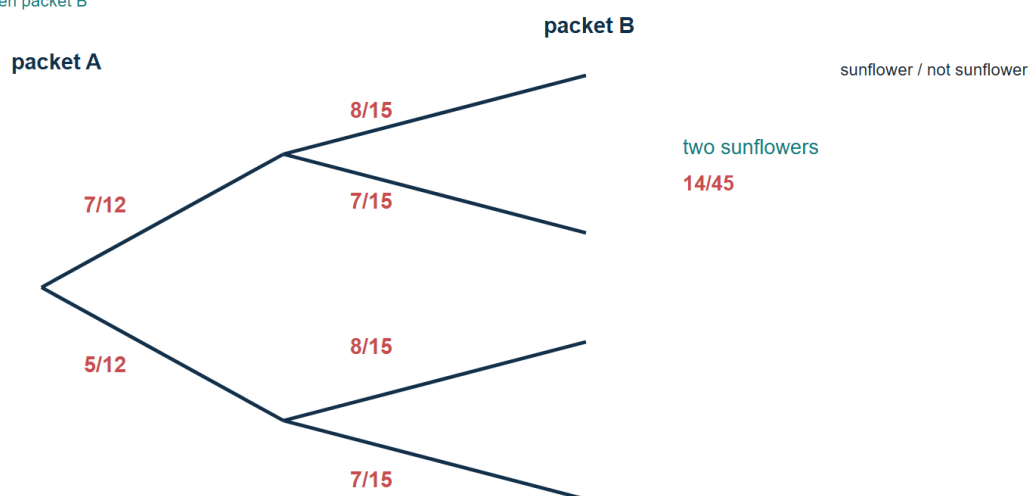
Chapter: Probability Diagrams - Venn & Tree Diagrams

MODULAR UNIT 1

2 marks

Q14 • Completed probability tree

Packet A then packet B



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Original question 13

4MA1-1H Q14 Q14(b)

ORIGINAL QUESTION

2 marks

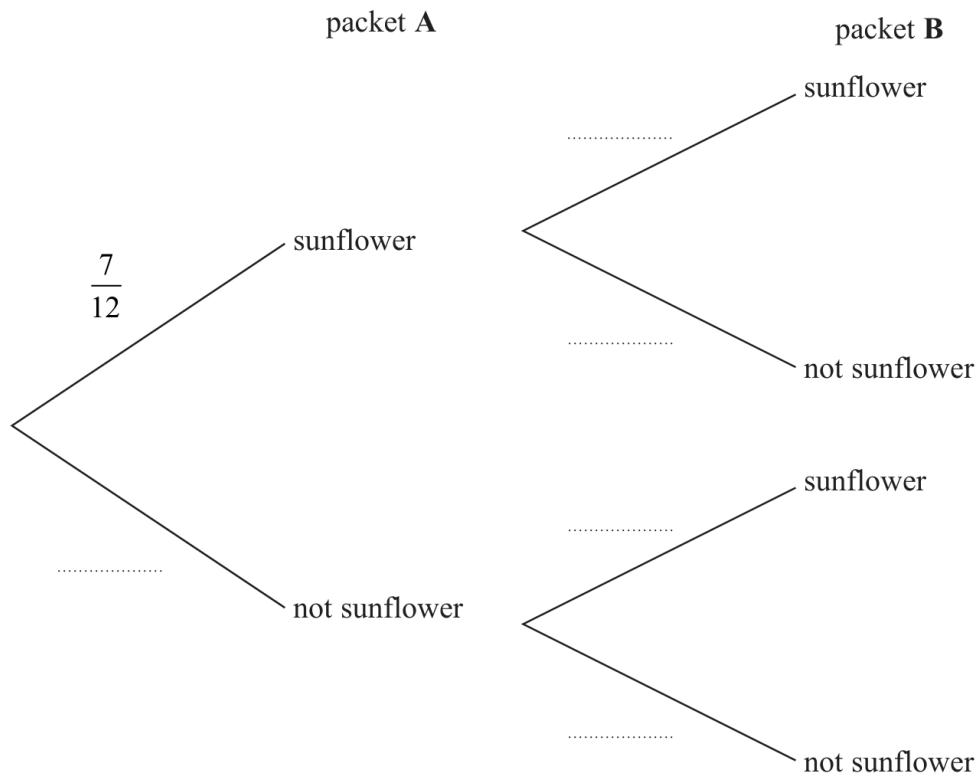
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There are 12 seeds in packet A and 7 of these are sunflower seeds.

There are 15 seeds in packet **B** and 8 of these are sunflower seeds.

Aika is going to take at random a seed from packet **A** and a seed from packet **B**

(a) Complete the probability tree diagram.



(b) Calculate the probability that Aika will take two sunflower seeds.

(2)

Worked solution 13

Chapter: Combined & Conditional Probability

MODULAR UNIT 1

2 marks

Q14(b) – Chapter: Combined & Conditional Probability

Family: Combine independent events • Variant: Find the probability that all independent events occur

Method

$$P(\text{two sunflowers}) = (7/12)(8/15).$$

*Why: Multiply the two independent packet probabilities.***Method**

$$P(\text{two sunflowers}) = 56/180 = 14/45.$$

*Why: Simplify the fraction.***Final answer**probability: $14/45$

Original question 14

4MA1-1H Q16

ORIGINAL QUESTION**4 marks**

16 The diagram shows a circle with centre O

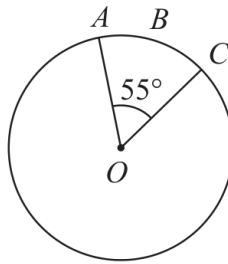


Diagram **NOT**
accurately drawn

A , B and C are points on the circle so that the length of the arc ABC is 5 cm.

Given that angle $AOC = 55^\circ$

work out the area of the circle.

Give your answer correct to one decimal place.

..... cm^2

Worked solution 14

Chapter: Circles, Arcs & Sectors

MODULAR UNIT 1

4 marks

Q16 – Chapter: Circles, Arcs & Sectors

Family: Use arc and sector measures • Variant: Recover radius or central angle from an arc/sector measure

Method

$$5 = (55/360) \times 2\pi r, \text{ so } r = 5 \times 360 / (55 \times 2\pi).$$

*Why: Use the arc-length formula $s = (\theta/360)2\pi r$.***Method**

$$r \approx 5.207 \text{ cm.}$$

*Why: Evaluate the radius.***Method**

$$\text{Area} = \pi r^2 \approx 85.2 \text{ cm}^2.$$

*Why: Use πr^2 .***Method**To 1 decimal place, the area is 85.2 cm^2 .*Why: Round as requested.***Final answer****area:** 85.2 cm^2

Original question 15

4MA1-1H Q18

ORIGINAL QUESTION**3 marks**

18 $A = w - \frac{x^2}{y}$

$w = 3.45$ correct to 2 decimal places.

$x = 1.9$ correct to 1 decimal place.

$y = 5$ correct to the nearest whole number.

Work out the lower bound of the value of A
Show your working clearly.

Worked solution 15

Chapter: Rounding, Estimation & Bounds

MODULAR UNIT 1

3 marks

Q18 – Chapter: Rounding, Estimation & Bounds

Family: Optimise a compound expression using bounds • Variant: Algebraic expression upper or lower bound

Method $w \geq 3.445$, $x < 1.95$ and $y \geq 4.5$.*Why: Use the correct bounds for the stated rounding.***Method**For a lower bound of $A = w - x^2/y$, use $w = 3.445$, $x = 1.95$ and $y = 4.5$.*Why: Minimise w and maximise x^2/y .***Method** $A = 3.445 - (1.95)^2/4.5 = 2.6$ (approximately).*Why: Substitute the bounds and calculate.***Final answer**

lower bound of A: 2.6

Original question 16

4MA1-1H Q20 Q20(a), Q20(b)

ORIGINAL QUESTION**4 marks**

20 (a) Express $7 + 12x - 3x^2$ in the form $a + b(x + c)^2$ where a , b and c are integers.

.....
(3)

C is the curve with equation $y = 7 + 12x - 3x^2$

The point *A* is the maximum point on **C**

(b) Use your answer to part (a) to write down the coordinates of *A*

(.....,)
(1)

Worked solution 16

Chapter: Completing the Square

MODULAR UNIT 1

4 marks

Q20(a) – Chapter: Completing the Square

Family: Complete the square for a non-monic quadratic • Variant: Negative leading coefficient or maximum form

Method

$$7+12x-3x^2=7-3(x^2-4x).$$

Why: Factor -3 from the quadratic terms.

Method

$$x^2-4x=(x-2)^2-4.$$

Why: Complete the square.

Method

$$\text{Therefore } 7+12x-3x^2=19-3(x-2)^2.$$

Why: Substitute the completed square and simplify.

Final answer

completed square: $19-3(x-2)^2$

Q20(b) – Chapter: Completing the Square

Family: Complete the square for a non-monic quadratic • Variant: Negative leading coefficient or maximum form

Method

The maximum occurs when $(x-2)^2=0$, giving $A=(2,19)$.

Why: The negative coefficient means the squared term is subtracted from 19.

Final answer

maximum point: $(2,19)$

Original question 17

4MA1-1H Q21

ORIGINAL QUESTION**5 marks**

- 21 The diagram shows the prism $ABCDEFGHJK$ with horizontal base $AEFG$

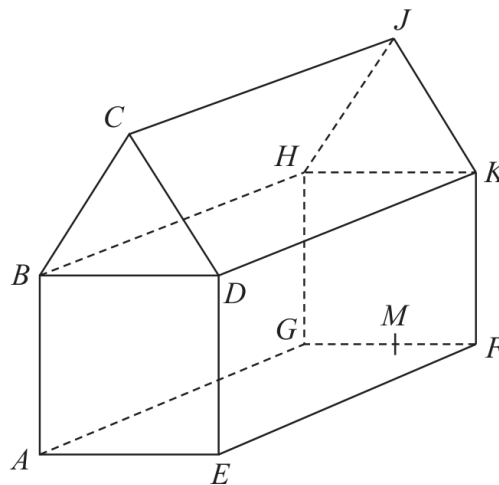


Diagram **NOT**
accurately drawn

$ABCDE$ is a cross section of the prism where
 $ABDE$ is a square
 BCD is an equilateral triangle

$$EF = 2 \times AE$$

M is the midpoint of GF so that JM is vertical.

$$\text{Angle } MAJ = y^\circ$$

$$\text{Given that } \tan y^\circ = T$$

find the value of T , giving your answer in the form $\frac{\sqrt{p} + \sqrt{q}}{17}$ where p and q are integers.

$T = \dots\dots\dots$

Worked solution 17

Chapter: 3D Pythagoras & Trigonometry

MODULAR UNIT 1

5 marks

Q21 – Chapter: 3D Pythagoras & Trigonometry

Family: Resolve multi-stage solid geometry • Variant: Use midpoint or symmetry structure in a 3D model

Method

$$AM = \sqrt{(x^2 + (4x)^2)} = x\sqrt{17}.$$

Why: Use Pythagoras in the horizontal base triangle.

Method

$$\text{The equilateral-triangle height is } \sqrt{((2x)^2 - x^2)} = x\sqrt{3}.$$

Why: Use the altitude of the equilateral cross-section.

Method

$$\tan y = (x\sqrt{3} + 2x) / (x\sqrt{17}) = (\sqrt{3} + 2) / \sqrt{17}.$$

Why: Use opposite over adjacent for angle MAJ.

Method

$$\text{Rationalising gives } (\sqrt{3} + 2)\sqrt{17}/17 = (\sqrt{51} + 2\sqrt{17})/17.$$

Why: Multiply numerator and denominator by $\sqrt{17}$.

Method

$$\text{Since } 2\sqrt{17} = \sqrt{68}, T = (\sqrt{51} + \sqrt{68})/17.$$

Why: Write the answer in the requested surd form.

Final answer

$$T: (\sqrt{51} + \sqrt{68})/17$$

Original question 18

4MA1-2H Q1 Q1(a)

ORIGINAL QUESTION**2 marks**

- 1 (a) Expand and simplify $(y + 4)(2 - y)$

.....

(2)

Worked solution 18

Chapter: Expanding Brackets

MODULAR UNIT 1

2 marks

Q1(a) – Chapter: Expanding Brackets

Family: Expand two linear expressions • Variant: Opposite signs, repeated bracket or square

Method

$$(y+4)(2-y)=2y+8-y^2-4y.$$

*Why: Expand both products.***Method**Collecting terms gives $8-2y-y^2$.*Why: Combine the y terms.***Final answer**expanded expression: $8-2y-y^2$

Original question 19

4MA1-2H Q1 Q1(b)

ORIGINAL QUESTION**2 marks**

(b) Factorise fully $15b^5c - 35b^3c^9$

(2)

Worked solution 19

Chapter: Factorising

MODULAR UNIT 1

2 marks

Q1(b) – Chapter: Factorising

Family: Extract an algebraic common factor • Variant: Greatest monomial common factor

Method

$$15b^5c - 35b^3c^9 = 5b^3c(3b^2 - 7c^8).$$

*Why: Take out the greatest common factor $5b^3c$.***Method**

Expanding the bracket checks the original expression.

*Why: Verify both terms after factorising.***Final answer****factorisation:** $5b^3c(3b^2 - 7c^8)$

Original question 20

4MA1-2H Q2

ORIGINAL QUESTION**3 marks**

2 Show that $6\frac{3}{4} \div 2\frac{4}{7} = 2\frac{5}{8}$

Worked solution 20

Chapter: Fractions

MODULAR UNIT 1

3 marks

Q2 – Chapter: Fractions

Family: Multiply or divide fractions and mixed numbers • Variant: Divide fractions or mixed numbers

Method $6\frac{3}{4} = \frac{27}{4}$ and $2\frac{4}{7} = \frac{18}{7}$.*Why: Convert both mixed numbers to improper fractions.***Method** $(\frac{27}{4}) \div (\frac{18}{7}) = (\frac{27}{4}) (\frac{7}{18}) = \frac{189}{72}$.*Why: Invert and multiply by the second fraction.***Method** $\frac{189}{72} = \frac{21}{8} = 2\frac{5}{8}$.*Why: Simplify and convert back to a mixed number.***Final answer**value: $2\frac{5}{8}$

Original question 21

4MA1-2H Q4 Q4(a)

ORIGINAL QUESTION**1 marks**

- 4 Each side of a regular octagon has a length of 18 mm, correct to the nearest 0.5 mm

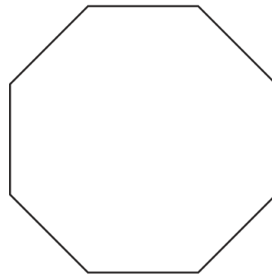


Diagram **NOT**
accurately drawn

- (a) Write down the lower bound of the length of each side of the octagon.

..... mm
(1)

Worked solution 21

Chapter: Rounding, Estimation & Bounds

MODULAR UNIT 1

1 marks

Q4(a) – Chapter: Rounding, Estimation & Bounds

Family: Recover single-measurement bounds • Variant: Direct lower and upper bounds

Method18 mm correct to the nearest 0.5 mm has lower bound $18 - 0.25 = 17.75$ mm.*Why: Half the rounding interval is 0.25 mm.***Final answer****lower bound:** 17.75 mm

Original question 22

4MA1-2H Q4 Q4(b)

ORIGINAL QUESTION**1 marks**

(b) Write down the upper bound of the length of each side of the octagon.

..... mm

(1)

Worked solution 22

Chapter: Rounding, Estimation & Bounds

MODULAR UNIT 1

1 marks

Q4(b) – Chapter: Rounding, Estimation & Bounds

Family: Recover single-measurement bounds • Variant: Direct lower and upper bounds

MethodThe upper bound is $18 + 0.25 = 18.25$ mm.*Why: Add half the rounding interval.***Final answer****upper bound:** 18.25 mm

Original question 23

4MA1-2H Q6

ORIGINAL QUESTION**5 marks**

- 6 A bag contains only pink sweets, white sweets, green sweets and red sweets.

The table gives each of the probabilities that, when a sweet is taken at random from the bag, the sweet will be green or the sweet will be red.

Sweet	pink	white	green	red
Probability			0.2	0.35

The ratio

number of pink sweets : number of white sweets = 2 : 1

There are 28 red sweets in the bag.

Work out the number of white sweets in the bag.

Worked solution 23

Chapter: Probability Toolkit

MODULAR UNIT 1

5 marks

Q6 – Chapter: Probability Toolkit

Family: Calculate theoretical single-event probability • Variant: Use category ratios in a probability model

MethodThere are 28 red sweets with probability 0.35, so total sweets $= 28 / 0.35 = 80$.*Why: Frequency equals probability times total.***Method**Pink plus white probability $= 1 - 0.2 - 0.35 = 0.45$.*Why: All four probabilities sum to 1.***Method**With pink:white=2:1, white probability is $0.45 / 3 = 0.15$.*Why: Split the 0.45 in the ratio 2:1.***Method**White sweets $= 80 \times 0.15 = 12$.*Why: Multiply the probability by the total.***Method**

Therefore the bag contains 12 white sweets.

*Why: State the requested count.***Final answer**

white sweets: 12

Original question 24

4MA1-2H Q14

ORIGINAL QUESTION**3 marks**

- 14 A zip wire is shown as the dashed line AC in the diagram.

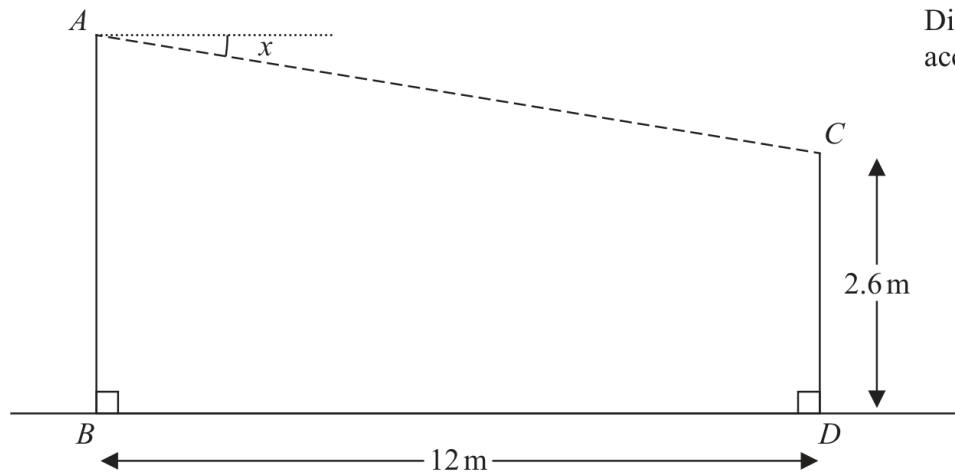


Diagram **NOT**
accurately drawn

The zip wire is supported by two vertical posts AB and CD standing on horizontal ground.

$$CD = 2.6 \text{ m} \quad BD = 12 \text{ m}$$

The zip wire makes an angle x with the horizontal, as shown in the diagram.
The design of the zip wire requires the angle x to be at least 5°

Work out the least possible height of the post AB
Give your answer correct to 3 significant figures.

..... m

Worked solution 24

Chapter: Right-Angled Triangles - Pythagoras & Trigonometry

MODULAR UNIT 1

3 marks

Q14 – Chapter: Right-Angled Triangles - Pythagoras & Trigonometry

Family: Resolve multi-step two-dimensional right-triangle models • Variant: Use a right-triangle result in a practical threshold or cost

Method

The vertical height difference is $y = 12 \tan 5^\circ = 1.04986... \text{ m}$.

Why: Use the right triangle between the posts.

Method

$AB = 2.6 + y = 3.64986... \text{ m}$.

Why: Add the height of CD.

Method

The least possible height is 3.65 m (3 s.f.).

Why: Round the result.

Final answer

AB: 3.65 m

Original question 25

4MA1-2H Q17

ORIGINAL QUESTION**3 marks**

17 Show that $\frac{\sqrt{12}}{\sqrt{3} + 2}$

can be written in the form $a + \sqrt{b}$ where a and b are integers.

Worked solution 25

Chapter: Surds

MODULAR UNIT 1

3 marks

Q17 – Chapter: Surds

Family: Rationalise a surd denominator • Variant: Binomial surd denominator using a conjugate

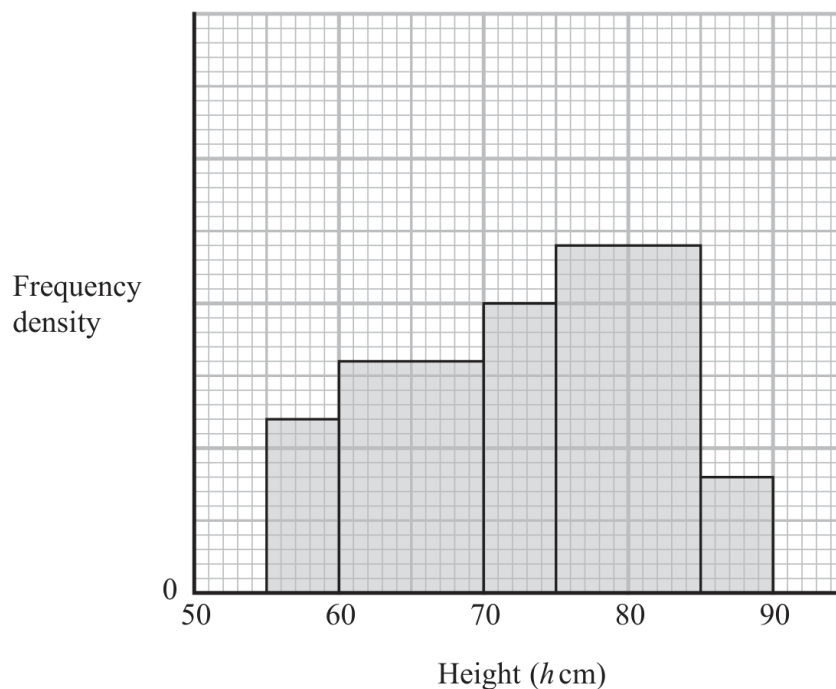
Method $\sqrt{12}/(\sqrt{3}+2) \times (\sqrt{3}-2)/(\sqrt{3}-2)$ rationalises the denominator.*Why: Multiply by the conjugate.***Method**The numerator is $6-2\sqrt{12}$ and the denominator is -1 .*Why: Expand and simplify the denominator.***Method**The result is $-6+\sqrt{48}$.*Why: Divide by -1 and write in the required form.***Final answer****surd form:** $-6+\sqrt{48}$

Original question 26

4MA1-2H Q20

ORIGINAL QUESTION**4 marks**

20 The histogram gives information about the heights, h cm, of some tomato plants.



There are 12 tomato plants for which $75 < h \leq 85$

One of the tomato plants is selected at random.

Find an estimate for the probability that this tomato plant has a height greater than 82.5 cm

Worked solution 26

Chapter: Histograms

MODULAR UNIT 1

4 marks

Q20 – Chapter: Histograms

Family: Interpret histogram areas • Variant: Estimate a histogram proportion, fraction or percentage

Method

The frequency densities are 0.6, 0.8, 1.0, 1.2 and 0.4, giving frequencies 3, 8, 5, 12 and 2.

*Why: Frequency equals class width multiplied by density.***Method**Total frequency = $3 + 8 + 5 + 12 + 2 = 30$.*Why: Add all class frequencies.***Method**Above 82.5 cm: half of the 75–85 class contributes $2.5 \times 1.2 = 3$, plus the 85–90 class contributes 2.*Why: Use the histogram area proportion.***Method**Probability = $5/30 = 1/6$.*Why: Divide the favourable frequency by 30.***Final answer**probability: $1/6$

Original question 27

4MA1-2H Q23

ORIGINAL QUESTION**6 marks**

23 The diagram shows triangle PQR

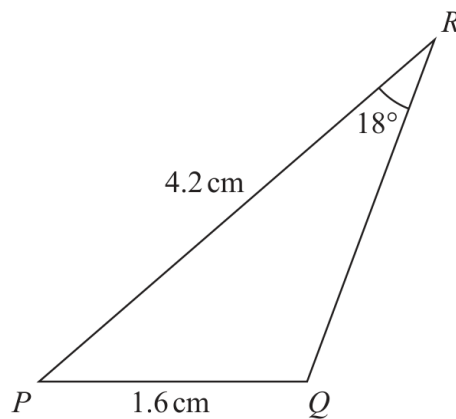


Diagram **NOT**
accurately drawn

$$PQ = 1.6 \text{ cm}$$

$$PR = 4.2 \text{ cm}$$

$$\text{Angle } PRQ = 18^\circ$$

Given that angle PQR is obtuse,

work out the area of triangle PQR

Give your answer correct to 3 significant figures.

..... cm^2

Worked solution 27

Chapter: Sine, Cosine Rule & Area of Triangles

MODULAR UNIT 1

6 marks

Q23 – Chapter: Sine, Cosine Rule & Area of Triangles

Family: Use the sine rule • Variant: Resolve the ambiguous sine-rule angle

MethodUsing the sine rule, $\sin Q/4.2 = \sin 18^\circ/1.6$, giving $Q = 54.2^\circ$ or 125.8° .*Why: The sine equation has an acute and an obtuse possibility.***Method**The question says Q is obtuse, so $Q = 125.8^\circ$.*Why: Select the permitted branch.***Method** $P = 180 - 125.8 - 18 = 36.2^\circ$.*Why: Angles in a triangle total 180° .***Method**The corresponding side RQ is 3.0585... cm.*Why: Use the cosine rule or the sine rule after finding P .***Method** $\text{Area} = \frac{1}{2} \times 4.2 \times 1.6 \times \sin 36.2^\circ$.*Why: Use two sides and the included angle.***Method** $\text{Area} \approx 1.98 \text{ cm}^2$ (3 s.f.).*Why: Round the calculator value.***Final answer****area:** 1.98 cm^2